Longitudinal care, without losing trust

A local-first workflow that keeps source, authority and action visible.

PROBLEM

The signal is buried in the record

Longitudinal notes accumulate across consultations, calls and patient reports. High-risk context, recent changes, open work and contradictions are easy to miss. A generic AI summary may look concise while hiding provenance, review state or role boundaries.

PRODUCT PRINCIPLE

WHAT MATTERS NOW

Care Glance ranks deterministic clinical state.

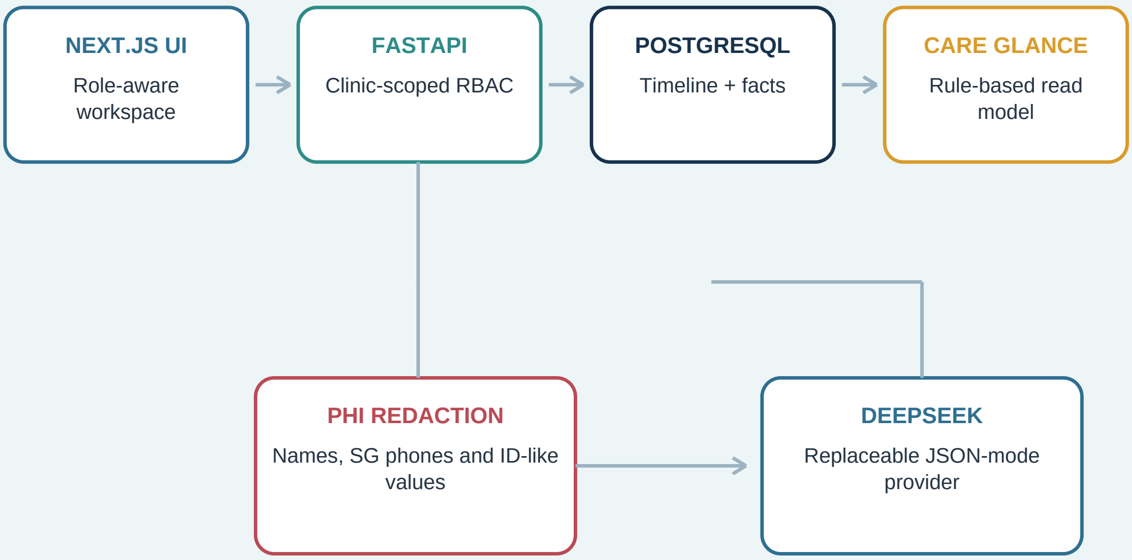
WHERE IT CAME FROM

Every highlight resolves to an exact source quote.

WHO VERIFIED IT

Review, revision and action remain attributable.

ARCHITECTURE



Boundary: raw transcripts stay in the database. The provider receives only redacted text. Care Glance never invokes the LLM on read.

Evidence before inference

The system separates source records, extracted facts, review state and action.

DATA MODEL

Longitudinal record

Clinic -> Patient -> ConsultSession -> Entry

Entry is the human-readable timeline unit. ClinicalFact carries structured value, risk, review status and an exact source span. Task and Conflict preserve actionable state. Highlight references one ClinicalFact. EntryVersion stores immutable full snapshots; revert appends a new version instead of erasing history.

Trust and provenance

Highlight -> ClinicalFact -> Entry -> ConsultSession

Every Glance item includes the exact source quote and occurrence time. AI suggestions remain suggested until a clinician accepts or rejects them. Audit events contain identifiers and state transitions - never full clinical text. Stale edits return 409 through optimistic concurrency.

AI PIPELINE

1. **STORE** raw locally →
2. **REDACT** deterministic PHI →
3. **GENERATE** DeepSeek JSON →
4. **VALIDATE** schema + quote →
5. **COMMIT** all-or-nothing

ROLE-BASED ACCESS

Cross-clinic resources return 404. UI hiding is not authorization. All 13 public Supabase tables use RLS with no client policy in v1: deny by default.

PATIENT

Own scope; accepted Glance; AI-patient scribe

STAFF

Clinic timeline; staff notes; internal comments

CLINICIAN

Review, conflict resolution, clinician notes

ADMIN

Read-only in v1

Bounded learning, deterministic risk

Adapt ranking carefully; never let feedback rewrite clinical truth.

IMPORTANCE AND SELF-LEARNING

Explainable score composition

final score = risk + recency + entity + open task + source authority + persistent critical + learning

The LLM never decides final ranking. A persistent critical allergy remains visible regardless of age. The clinician interface presents readable reasons, not raw scores.

Learning bonus = clamp(accept_count x 0.25 - reject_count x 0.20, 0, 3)
Feedback transfers only across similar entities inside the same clinic and never changes risk labels.

EXPLAINABLE DATA DECAY

NEVER DECAY

Allergies and clinician-confirmed critical facts

FULL FIDELITY

Recent facts plus persistent or medium/high-risk context

CANDIDATE ONLY

Old, low-risk, transient context; no deletion in v1

PERFORMANCE EVIDENCE

4.35 ms	4.87 ms	5.23 ms	<= 300 ms
P50 LOCAL WARM PATH	P95 LOCAL WARM PATH	MAX	TARGET

20 warmups + 200 measured requests using FastAPI TestClient, local SQLite and fixed synthetic data. This is a regression baseline, not a public-network Supabase SLA.

TRADE-OFFS

Purposeful constraints for a 72-hour build

Prototype identity: X-Demo-User-ID demonstrates server-side policy but is not production authentication. **Revision:** full snapshots favor simple, reliable diff and append-only revert. **Conflict scope:** v1 detects only same-medication, different-dose discrepancies. **RLS:** deny-by-default supports a backend-only boundary; client policies require a separate design. **Scope:** no voice capture, real patient data or automated treatment advice.